Summary in Graph

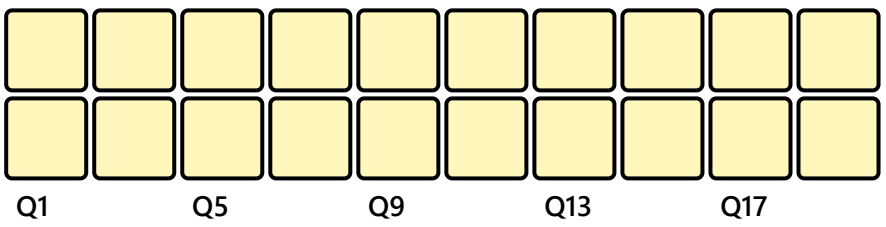
Exam Summary (GO Classes GATE CS | Calculus | Weekly Quiz 3)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	20 (10 + 10)
Total Marks:	30 (10 + 20)
Exam Duration:	60 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (12) out of 121 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Calculus

The intervals for which the function $F(x) = x^4 - 4x^3 + 4x^2 + 6$ increases are

A. $x < 0, 1 < x < 2$

B. only $x > 2$

C. $0 < x < 1, x > 2$

D. only $0 < x < 1$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 01sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Let them x, y be positive real numbers such that $xy = 1$. What is the minimum value$x + y$?

A. $\sqrt{2}$
B. 1
C. 2
D. The minimum value does not exist

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Let $f, g : [a, b] \to \mathbb{R}$ be continuous functions. Define

$$h(x) = \min\{f(x), g(x)\}$$

Which of the following statements is **true**?

A. h is not necessarily continuous on $[a, b]$

B. h is continuous only if $f(x) \leq g(x)$ for all $x \in [a, b]$

C. h is continuous on $[a, b]$

D. h is continuous only at points where $f(x) = g(x)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Let

$$f(x) = \begin{cases} -x + c & \text{if } x \leq 1 \\ 6 - 2x^2 & \text{if } x > 1 \end{cases}$$

Find the value of c such that $f(x)$ is continuous at $x = 1$.

A. 2
B. 4
C. 5
D. 6

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 Numerical Type | Award: 1 | Penalty: 0 | Calculus

Your friend has two functions, $f(x)$ and $g(x)$. He calls their product $h(x)$; i.e., $h(x) = f(x)g(x)$. One day he asks you to compute $g'(5)$, but only tells you

- $g(5) = 1$,
- $f(5) = 2$,
- $f'(5) = 3$,
- $h'(5) = 4$.

What is $g'(5)$?

Your Answer: | Correct Answer: 0.5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Evaluate the limit:

$$\lim_{x \to 3} \frac{x^2 - x - 6}{x^2 - 9}$$

A. $\frac{5}{6}$
B. 0
C. ∞
D. $\frac{1}{2}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

What is the derivative of $g(x) = \frac{1}{f(\sqrt{x})}$?

A. $\frac{f'(\sqrt{x})}{\sqrt{x} f(\sqrt{x})^2}$

B. $\frac{-f'(\sqrt{x})}{\sqrt{x} f(\sqrt{x})^2}$

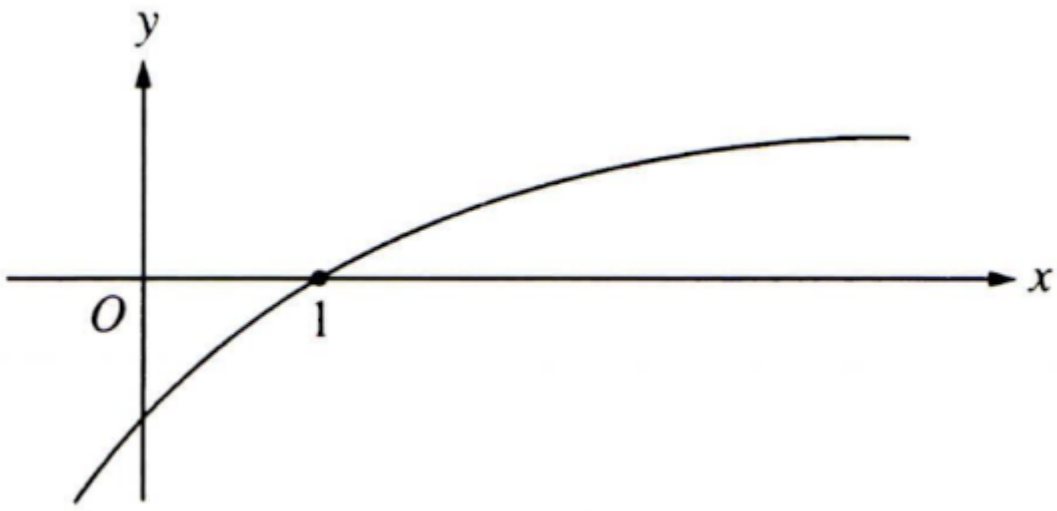
C. $\frac{f'(\sqrt{x})}{2\sqrt{x} f(\sqrt{x})^2}$

D. $\frac{-f'(\sqrt{x})}{2\sqrt{x} f(\sqrt{x})^2}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

The graph of a twice-differentiable function f is shown in the figure below. Which of the following is true?



A. $f(1) < f'(1) < f''(1)$

B. $f(1) < f''(1) < f'(1)$

C. $f'(1) < f(1) < f''(1)$

D. $f''(1) < f(1) < f'(1)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

x	0	2	5	9	11
$g(x)$	1	2.8	1.7	1	3.4

The table above shows selected values of a continuous function **g.** For $0 \leq x \leq 11$, what is the fewest possible number of times $g(x) = 2$?

A. One
B. Two
C. Three
D. Four

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

A function $f(x)$ is known to satisfy $f(0) = 1.05$ and $f'(0) = -0.1$. Using the tangent line approximation, what is a reasonable estimate for it $f(0.1)$?

A. 0.95
B. 1.00
C. 1.04
D. 1.10

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Let f be a continuous and one-to-one $(1-1)$ function on the interval $[0,1]$, and suppose $f(0) < f(1)$. Which of the following must be true for every $x \in (0,1)$?

A. $f(x) = f(0)$ or $f(x) = f(1)$

B. $f(x) \notin [f(0), f(1)]$

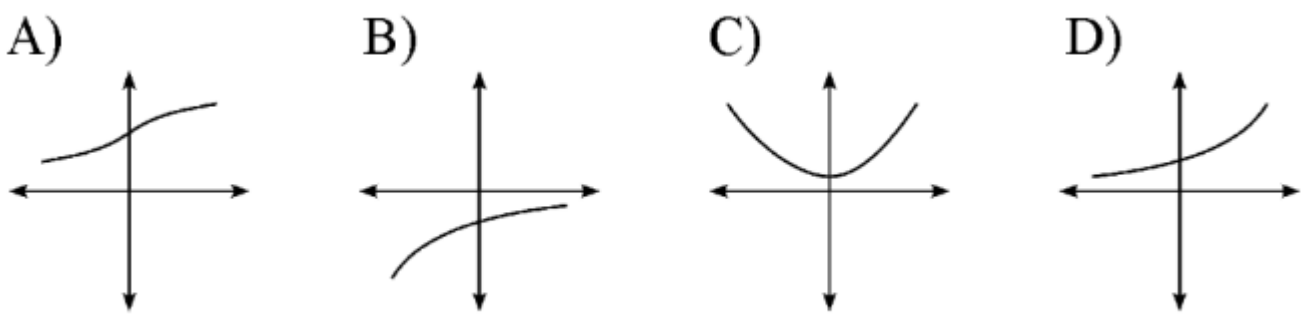
C. $f(0) < f(x) < f(1)$

D. $f(x) > f(1)$ or $f(x) < f(0)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

$f(x)$, $f'(x)$, and $f''(x)$ are all positive for any real number x. Which of the following graphs could be a graph of f ?



Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

What are all values of x for which the function $f(x) = x^3 + 6x^2 + 9x + 1$ is increasing?

A. $(-\infty, -3)$only

B. $(-3, -1)$only

C. $(-1, \infty)$only

D. $(-\infty, -3) \cup (-1, \infty)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

If g is a differentiable function such that $g(x) < 0$ for all real numbers x and if $f'(x) = \left(x^2 - 4\right) g(x)$, which of the following is true?

A. f has a relative maximum at $x = -2$ and a relative minimum at $x = 2$.

B. f has a relative minimum at $x = -2$ and a relative maximum at $x = 2$.

C. f has a relative minima at $x = -2$ and at $x = 2$.

D. f has a relative maxima at $x = -2$ and at $x = 2$.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Numerical Type | Award: 2 | Penalty: 0 | Calculus

Evaluate the limit

$$\lim_{x \to \infty} (\sqrt{x+1} - \sqrt{x+2})$$

Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

If $\lim_{x \to a} f(x) = 0$ and $\lim_{x \to a} g(x) = 0$ then

$$\lim_{x \to a} \frac{f(x)}{g(x)}$$

A. 1
B. 0
C. does not exist
D. Not enough information is given to say anything about the limit.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Let $f : \mathbb{R} \to \mathbb{R}$ be a twice continuously differentiable function such that

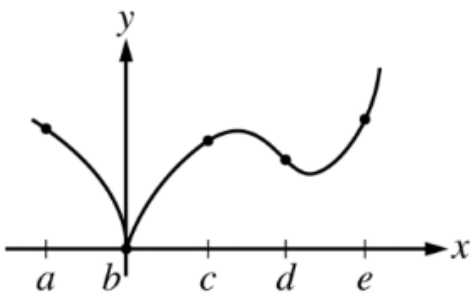
$$f(0) = f(1) = f'(0) = 0.$$

Which of the following statements is necessarily true?

A. $f''(x) = 0$ for all $x \in \mathbb{R}$
B. $f''(0) = 0$
C. There exists some $x \in (0, 1)$ such that $f''(x) = 0$
D. $f''(x) \neq 0$ for any $x \in (0, 1)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus



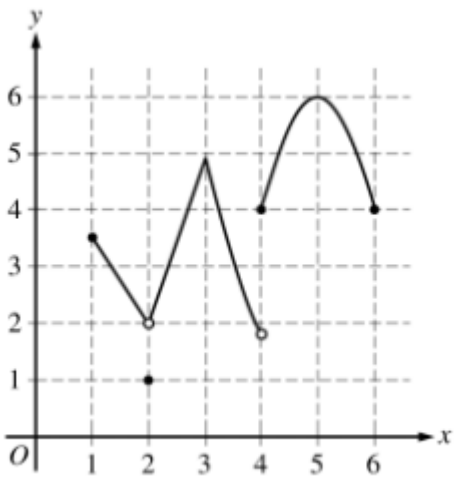
Graph of f

The graph of the function f is shown in the figure above. For which of the following values of x is $f'(x)$ positive and increasing?

A. a
B. b
C. c
D. e

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus



Graph of f

The graph of the function f is shown above. Which of the following statements is false?

A. $\lim_{x\to 2} f(x)$ exists.

B. $\lim_{x\to 3} f(x)$ exists.

C. $\lim_{x\to 4} f(x)$ exists.

D. The function f is continuous at $x = 3$.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Let $f : [0, 5] \to \mathbb{R}$ be continuous function with a maximum at $x = 2$ then

A. the derivative of f at 2 may not exist
B. the derivative of f at 2 must exist and be nonzero
C. the derivative of f at 2 must exist and be zero
D. the derivative of f at 2 can not exist

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

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